



Fig. 3: A classic IoT Things architecture

and M2M solutions. For telecom service providers, this essentially means that they create a layer of a network management framework to configure and manage the network for IoT solutions. In Fig. 2, the layer below Applications shows what functionalities they implement in their network in order to be able to offer IoT services.

IoT users and designers need to understand the interconnection options and use these effectively. Interconnections offered can range from a simple GPRS/SMS based solution to a sophisticated 4G (LTE) based solution. Designers need to understand and estimate the data that the Things will be sending and receiving, and decide on the type of interface they will design for the Things.

Also, service providers offer varied quality of services (QoS) with differentiated pricing and these have an impact on the design of Things in terms of battery power, memory resources and CPU power needed in the design.

Applications. Finally, central applications play a vital role in the IoT offering. Designers of Things need to understand the architecture of the applications. Many a time, designers finalise the design and realise that the applications may need more features than what was originally envisaged. It is important to have a clear overview of end-to-end solutions. This is critical when applications are mission-critical like security, healthcare and so on.

Hardware architecture for the IoT

IoT system architectures are not completely new; all existing microcontrollers (MCUs) can be used. However, choice of the MCU is dependent on the specific on-chip resources and the power needed for the application. This is in addition to the suitability of the MCU for the sensor being used. Device vendors are now integrating communication interfaces into the MCUs so that most Things can be designed out of a single processor/controller.

However, some aspects that a designer must watch out for in the hardware designs are:

1. Data that needs to be captured
2. Volume of data to be sent
3. Frequency of data transmission
4. Communication interface type
5. Type of actuators controlled

These parameters decide the components in the MCU, which, in turn, decide the cost of hardware. Many a time, low-cost Things are not meant to be serviced. When these go faulty, these are discarded and replaced with new ones. This puts additional stress on cost and forces the designers to have a good overview of the end-to-end solution in order to be able to design an optimal hardware design.

Software architecture for the IoT

Most conventional architectures used for the IoT are based on open source components. A classic IoT Things

architecture is shown Fig. 3. Most IoT solutions use open source Linux as base, as this helps in software development faster without having to wait for the target hardware.

There are good standard IoT-centric frameworks now available, which reduce development time. What is unique in the software architecture is the Constrained Application Protocol (CoAP), which is unique to IoT applications and offers a common way to communicate with IoT devices.

Some ready-to-use IoT frameworks

Thread Group's Thread Framework uses existing IETF/IEEE standards (ZigBee and 6LowPAN). It is capable of handling up to 250 devices in a personal area network (PAN) or home area network (HAN). This framework, when used, needs a gateway to communicate to the central application server. (www.threadgroup.org)

Eclipse Open IoT Workgroup has two frameworks available. One uses Open Services Gateway Initiative (OSGi) called Kura OSGi and the other is Mibini framework, which is specially developed for Lua programming language. (www.iot.eclipse.org)

There is one from Open Interconnect Consortium called IoTivity. Linux Foundation hosts this. IoTivity uses profiles on the application side using four major building-blocks, namely, discovery, data transmission, data management and device management. One advantage of this framework is that it runs on mobile operating systems, too. (www.iotivity.org). Its architecture very closely resembles career-class solutions with a modular approach.

In addition to these, vendors like Linear Technology and Microsoft also offer IoT frameworks. These are managed and controlled by them, too. Even enterprise software vendors like Oracle offer IoT frameworks, which can seamlessly interwork with Oracle database. ●